

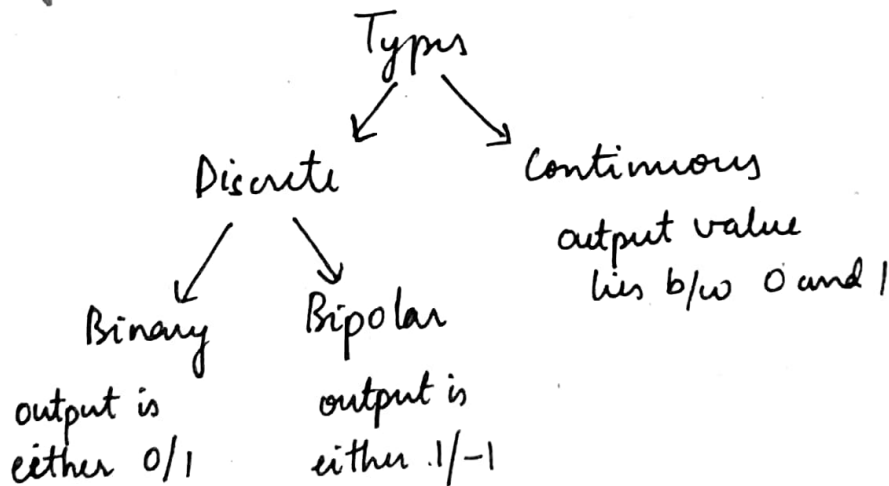
ASSIGNMENT 2

Perumalla Dhanan

AP21110010201

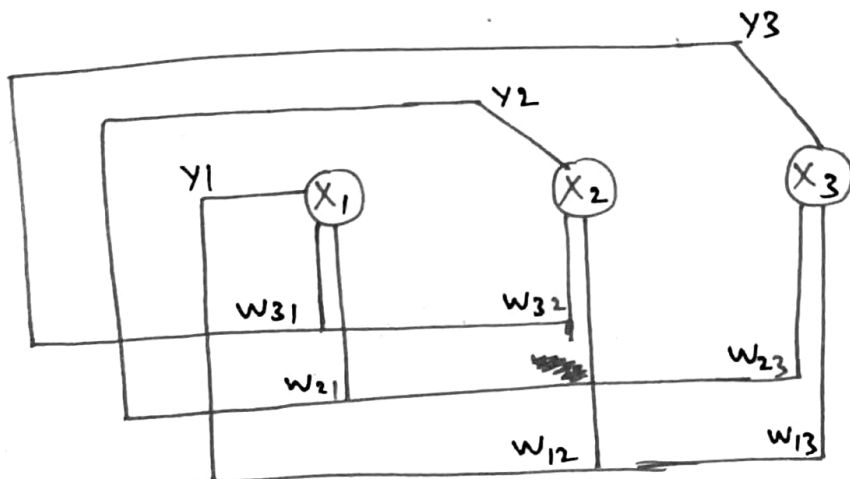
(Q1) Explain the concept of the hopfield network and how it functions as an associated memory neural network? How does the network reach a stable state?

(Ans) The hopfield network is a type of recurrent neural network, named after John Hopfield, primarily used as an associative memory system. It is designed to store and recall patterns, particularly used for pattern recognition tasks.



Architecture of Hopfield network

- Individual nodes preserve their states until required on update.
- The node to be update is selected randomly.
- Each node is connected to all other nodes except itself.
- The state of each node is either 0/1 or $1/-1$.
- It is symmetric in nature i.e. $w_{ij} = w_{ji}$



$X_1, X_2, X_3 \rightarrow$ Inputs

$Y_1, Y_2, Y_3 \rightarrow$ Outputs obtained from each node.

$w_{ij} \rightarrow$ Represents weight connected from i to j .

Hopfield as an associative memory

The Hopfield network is widely used for associative memory, which allows it to store multiple patterns and retrieve one of them when provided with a noisy or partial version of the stored pattern.

1. Training (Storing Patterns)

It is based on Hebbian principle, given in 1949. Weights are updated using specific rule which is different for binary & bipolar.

Binary -

$$W_{ij} = \sum_{n=1}^N (2x_i(n) - 1)(2x_j(n) - 1)$$

Bipolar -

$$W_{ij} = \sum_{n=1}^N (x_i(n) x_j(n))$$

Reaching a Stable State

- Asynchronous Update

Neurons are updated one at a time based on the weighted sum of inputs from other neurons. This asynchronous updating ensures that the network does not oscillate and instead gradually approaches a stable state.

- Convergence

The network eventually reaches a state where no further updates occur, meaning the energy function reaches a local minimum. This stable state represents one of the stored patterns, effectively allowing the network to recall the pattern from the initial noisy input.

Q2) Explain the fundamental difference in how ART1 and ART2 process input data. How do they differ between ART1 and ART2? Explain with diagrams.

Ans) DIFFERENCES BETWEEN ART1 & ART2

① Type of Input Data -

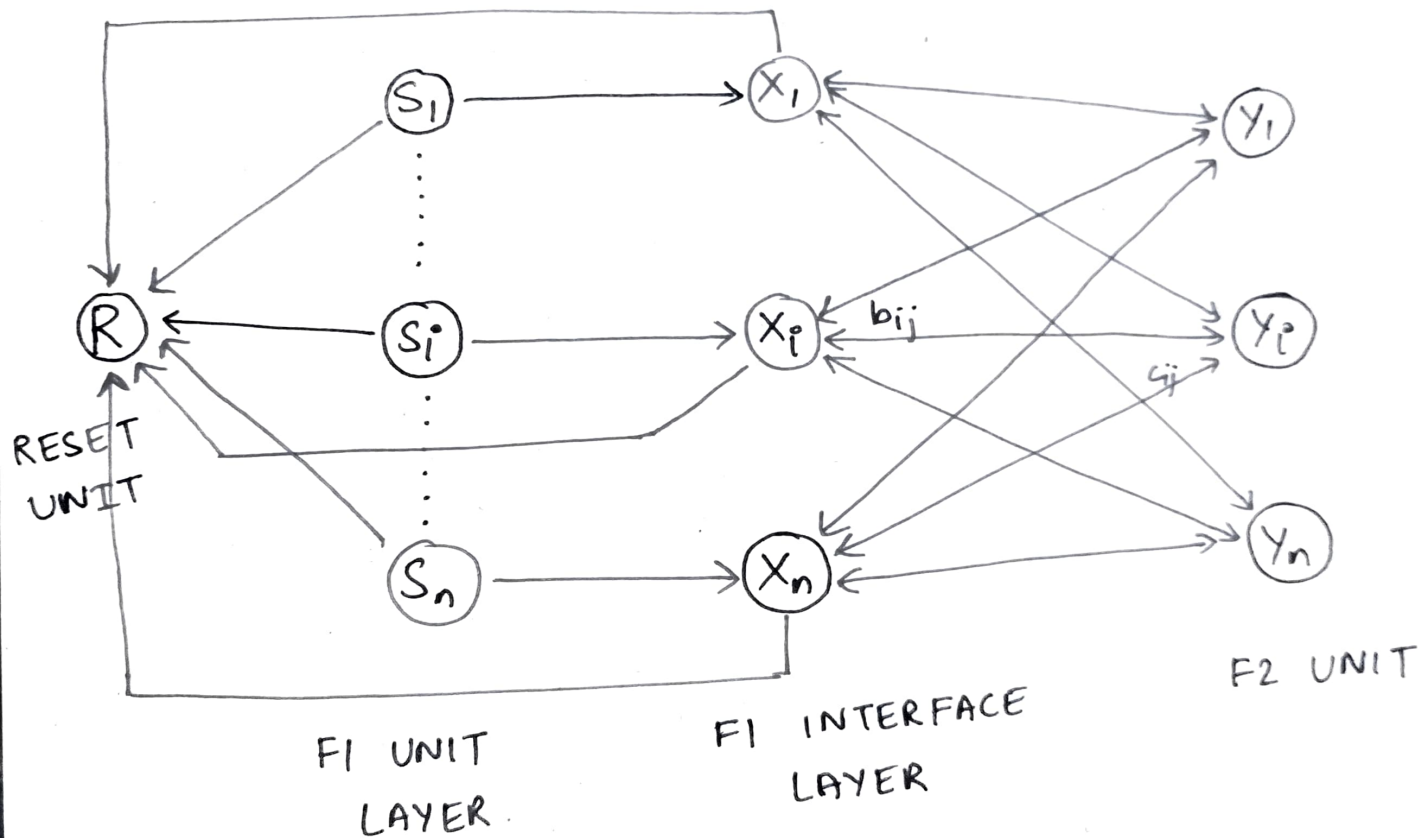
ART1 works with binary input patterns.

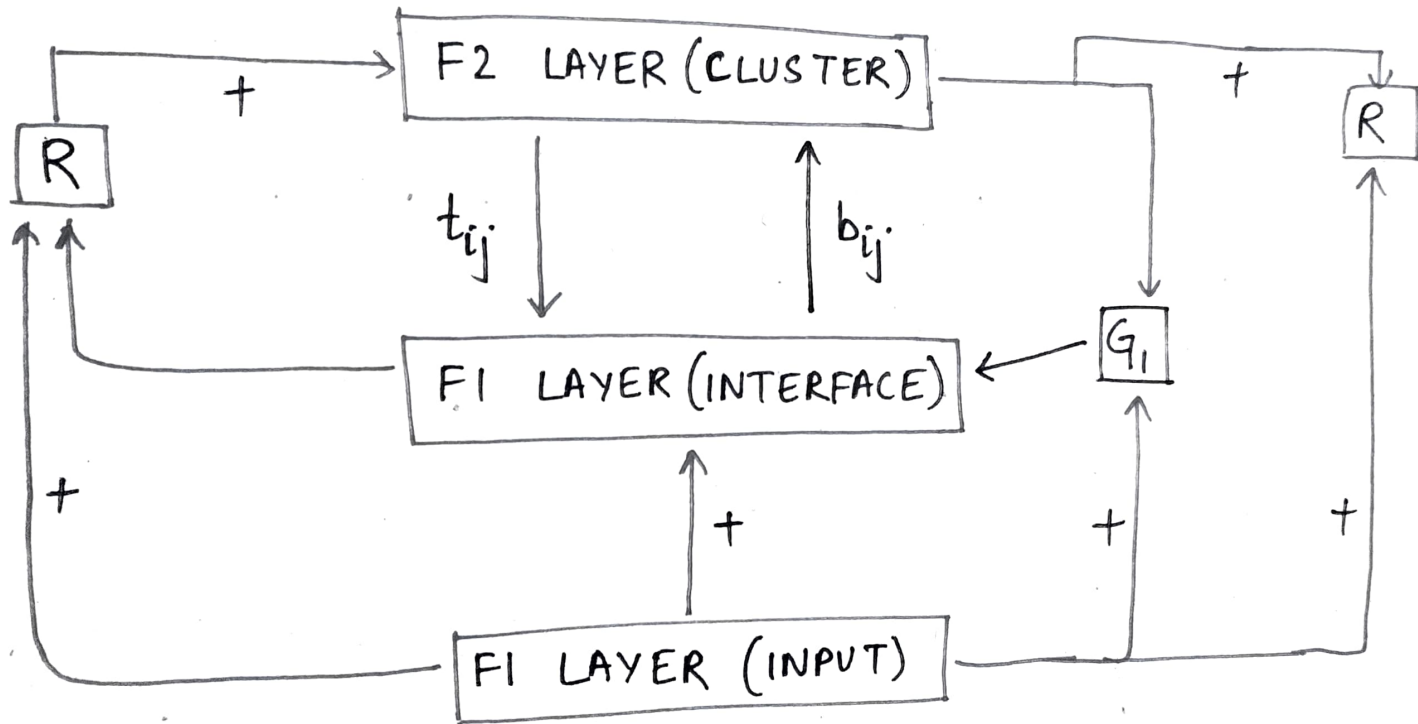
ART2 works with analog input patterns.

② Processing of Input Data -

Since ART1 deals with binary inputs, its operations are designed for simple bitwise comparisons.

In ART2, network handles continuous (analog) data which requires more complex operations such as vector normalization, noise suppression, etc.





ART 1

- ① Binary Input (0 & 1)
- ② Preprocessing - None
- ③ Simple Binary Comparisons
- ④ It is used for mainly clean, discrete data.
- ⑤ Vigilance test - Binary pattern matching
- ⑥ Less Complex.

ART 2

- ① Analog Input (Real Value)
- ② Preprocessing - Normalization
- ③ Complex Operations
- ④ Can handle noisy, continuous data.
- ⑤ Vigilance Test - Continuous pattern matching
- ⑥ More Complex

- * In ART 1, the F1 layer is simpler because it processes binary inputs. In ART 2, the F1 layer involves complex processing.
- * ART 1 performs direct binary matching whereas ART 2 uses continuous matching.
- * ART 2 requires additional steps for competitive learning to handle real valued data, making it even more

flexible. This allows ART2 to recognise patterns
with more variability than ART1.